

RESOLUTE BAY, NU, Canada

WMO: 719240

Lat: 74.720N

Lon: 94.970W

Elev: 222

StdP: 14.58

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-42.6	-39.8	-48.7	0.3	-42.1	-46.7	0.4	-39.4	38.1	-16.2	35.5	-14.5	9.1	330	1.204

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	53.6	47.4	50.1	45.2	47.0	43.2	47.7	53.1	45.4	49.6	43.4	46.7	11.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
43.6	42.2	49.6	41.7	39.2	47.6	40.1	36.8	45.7	19.1	53.3	17.9	49.9	16.9	46.8	55.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.5	30.7	27.6	DB	-46.0	57.6	3.9	5.0	-48.8	61.2	-51.0	64.1	-53.2	66.9	-56.0	70.6	(4)
(5)			WB	-45.9	50.7	3.5	2.7	-48.4	52.6	-50.5	54.2	-52.4	55.8	-55.0	57.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-23.5	-25.5	-20.9	-5.5	14.6	33.0	40.6	36.6	25.7	10.1	-5.4	-16.8		
	DBStd	25.15	8.73	8.74	11.23	9.76	8.89	5.92	5.60	5.67	6.05	9.95	10.36	10.39		
	HDD50	16278	2279	2115	2199	1665	1097	510	298	416	730	1237	1661	2071		
	HDD65	21748	2744	2535	2664	2115	1562	960	758	881	1180	1702	2111	2536		
	CDD50	6	0	0	0	0	0	0	6	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	12.3	11.1	10.9	11.4	11.6	11.9	12.1	12.4	12.6	14.2	14.0	13.3	11.4		
	PrecAvg	6.0	0.1	0.1	0.2	0.3	0.4	0.5	1.0	1.3	0.9	0.6	0.3	0.2		
(15) Precipitation	PrecMax	8.6	0.4	0.4	0.8	1.3	1.1	1.2	3.2	3.1	2.2	1.7	0.9	0.7		
	PrecMin	3.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0		
	PrecStd	1.3	0.1	0.1	0.2	0.3	0.3	0.3	0.7	0.7	0.5	0.3	0.2	0.2		
	PrecStd	1.3	0.1	0.1	0.2	0.3	0.3	0.3	0.7	0.7	0.5	0.3	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	11.5	2.1	14.3	20.6	34.5	52.0	60.0	54.3	40.6	30.7	22.2	17.2			
	0.4% MCWB	11.2	1.0	13.1	19.4	32.2	45.4	50.2	48.0	37.6	29.9	21.5	16.2			
	2% DB	-2.2	-3.3	6.6	16.4	31.9	46.9	55.9	50.4	36.8	28.4	17.7	8.4			
	2% MCWB	-3.0	-4.0	5.7	15.2	30.4	42.3	48.7	45.7	35.0	27.7	17.1	7.6			
	5% DB	-7.4	-8.4	1.4	12.7	29.7	43.4	52.7	47.5	34.8	26.6	13.9	3.1			
	5% MCWB	-8.0	-9.0	0.5	11.6	28.2	39.9	46.9	43.8	33.6	25.9	13.0	2.4			
	10% DB	-11.1	-12.5	-4.3	8.2	27.2	40.5	49.5	45.0	33.1	24.2	9.2	-1.4			
	10% MCWB	-11.6	-13.1	-5.1	7.1	25.8	37.8	44.8	42.1	32.1	23.4	8.4	-2.2			
	0.4% WB	10.6	1.1	13.0	19.8	33.1	46.0	51.5	48.3	37.7	30.3	21.4	16.3			
	0.4% MCDB	10.9	2.0	14.0	20.5	33.9	50.8	57.8	53.5	39.8	30.7	22.0	17.6			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-2.9	-4.0	5.8	15.2	30.5	42.4	49.1	46.0	35.3	27.9	17.0	7.7			
	2% MCDB	-2.2	-3.3	6.5	16.1	31.7	46.5	55.1	49.7	36.4	28.3	17.5	8.4			
	5% WB	-8.1	-8.9	0.5	11.6	28.3	40.0	47.0	44.1	33.8	26.0	13.1	2.5			
	5% MCDB	-7.6	-8.5	1.3	12.7	29.7	43.0	52.4	46.9	34.7	26.6	13.9	3.2			
	10% WB	-11.6	-13.0	-5.0	7.2	25.9	37.9	44.9	42.3	32.2	23.5	8.5	-2.1			
	10% MCDB	-11.1	-12.5	-4.2	8.2	27.2	40.2	49.2	44.7	32.9	24.2	9.1	-1.4			
	MDBR	10.0	9.9	10.5	10.7	9.1	6.8	8.8	7.0	6.1	8.8	9.9	10.5			
	5% DB MCDBR	13.3	13.7	12.1	12.2	8.9	10.0	13.4	10.8	6.3	7.7	11.7	13.2			
	5% DB MCWBR	13.1	13.4	12.0	11.7	8.1	7.1	8.7	7.6	5.4	7.9	11.5	12.9			
	5% WB MCDBR	13.3	13.4	12.1	12.3	8.4	9.5	13.1	10.3	6.0	7.8	11.7	13.2			
	5% WB MCWBR	13.1	13.2	12.0	11.8	7.8	7.0	9.0	7.6	5.4	8.0	11.6	12.9			
(40) Clear-Sky Solar Irradiance	taub	N/A	0.149	0.220	0.281	0.304	0.303	0.303	0.291	0.249	0.162	N/A	N/A			
	taud	N/A	1.877	2.174	2.089	2.127	2.296	2.470	2.523	2.419	1.914	N/A	N/A			
	Ebn at Noon	0	109	231	254	266	274	271	259	225	88	0	0			
	Edh at Noon	0	6	18	31	36	32	26	21	15	5	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	29	433	1296	2024	2132	1597	879	350	72	0	0			
	RadStd	0	4	26	51	104	223	175	108	47	13	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.22	+2.81	+1.80	N/A	N/A	N/A	-441	-439	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page